

ALTAY, China

WMO: 510760

Lat: 47.7410N

Lon: 88.0710E

Elev: 2418

StdP: 13.46

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.9	-16.7	-27.9	1.3	-21.7	-22.6	1.7	-16.2	12.1	10.2	9.8	8.4	2.6	0	0.516

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.9	89.3	65.0	86.6	63.8	83.9	62.8	67.8	83.6	66.2	81.7	64.7	79.9	5.1	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.4	92.3	74.7	60.6	86.3	73.0	58.9	81.3	71.8	33.7	83.6	32.3	81.5	31.1	80.0	78.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
17.3	14.3	12.0	-26.8	94.9	6.3	3.1	-31.3	97.2	-35.0	99.0	-38.5	100.8	-43.1	103.0		
			-27.0	71.6	6.2	2.8	-31.5	73.7	-35.2	75.3	-38.7	76.9	-43.2	79.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	41.6	3.8	10.0	26.4	48.1	60.0	69.4	71.5	68.7	58.2	45.0	26.4	9.7
	DBStd	25.88	10.48	10.37	13.10	10.17	8.40	6.23	5.36	6.21	8.21	9.15	12.31	9.95
	HDD50	5644	1434	1119	735	149	16	0	0	0	22	209	709	1251
	HDD65	9141	1899	1539	1197	508	200	29	8	32	232	622	1159	1716
	CDD50	2569	0	0	3	94	325	581	666	580	267	52	1	0
	CDD65	590	0	0	0	3	44	159	210	146	27	1	0	0
	CDH74	5289	0	0	0	22	380	1409	1855	1370	249	4	0	0
	CDH80	1636	0	0	0	2	80	442	626	437	49	0	0	0
Wind	WSAvg	4.7	2.8	3.3	4.5	6.9	6.8	5.6	4.7	4.9	5.0	4.9	4.1	2.9
Precipitation	PrecAvg	7.8	0.6	0.4	0.4	0.5	0.7	0.6	1.0	0.7	0.6	0.6	0.9	0.8
	PrecMax	13.3	3.0	2.0	2.1	1.7	2.2	2.2	3.3	2.4	2.2	1.6	2.6	2.5
	PrecMin	3.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.1
	PrecStd	2.4	0.6	0.4	0.4	0.4	0.5	0.4	0.7	0.6	0.5	0.4	0.6	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.7	34.7	61.1	77.0	86.5	91.0	94.0	92.3	85.4	73.3	56.9	34.4
		MCWB	27.9	29.5	45.9	53.0	59.0	64.7	69.6	65.6	59.8	50.0	42.4	29.6
	2%	DB	25.8	30.6	54.7	71.7	81.8	87.9	89.4	88.2	80.2	67.2	50.9	29.5
		MCWB	23.2	26.1	42.8	50.0	57.0	63.2	66.6	64.9	57.3	47.7	39.4	26.6
	5%	DB	21.6	27.1	49.3	67.4	78.2	85.1	86.5	84.8	76.5	62.9	46.2	26.4
		MCWB	19.3	23.5	40.0	48.2	55.4	62.2	65.5	63.4	55.3	46.6	37.5	24.3
	10%	DB	18.2	24.0	44.4	63.6	74.6	82.2	83.4	81.7	72.8	58.8	41.9	23.5
		MCWB	16.0	21.1	36.8	46.8	53.3	61.4	64.3	62.2	54.0	45.0	34.8	21.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	28.7	30.0	46.8	54.8	61.2	67.5	72.3	69.2	61.9	52.1	44.2	30.5
		MCDB	30.1	34.1	58.1	71.5	81.4	83.7	89.6	86.4	81.6	64.7	53.7	33.2
	2%	WB	23.2	26.6	43.5	52.0	58.6	65.7	69.0	66.7	59.2	49.7	40.6	26.9
		MCDB	25.5	30.2	53.2	66.6	77.3	82.6	84.8	83.2	76.1	62.9	49.4	28.8
	5%	WB	19.5	23.7	40.6	49.9	57.0	64.2	67.1	64.9	57.1	47.8	38.0	24.4
		MCDB	21.5	26.8	48.8	63.6	74.7	80.9	82.2	81.1	73.1	60.3	45.3	26.7
	10%	WB	15.9	21.1	37.2	48.1	55.2	62.7	65.6	63.3	55.1	46.0	35.3	21.1
		MCDB	18.1	23.9	43.9	60.9	71.1	78.5	80.0	78.8	69.8	57.3	41.2	23.4
Mean Daily Temperature Range	5% DB	MDBR	16.8	18.0	16.8	18.2	19.0	19.7	19.9	21.3	21.7	17.8	14.5	15.3
		MCDBR	16.8	19.3	19.0	22.2	22.0	22.6	23.6	24.5	24.9	22.6	18.4	15.2
		MCWBR	14.6	15.9	11.7	10.6	9.2	9.6	11.5	11.8	11.0	10.9	10.8	12.6
	5% WB	MCDBR	15.9	18.2	17.7	19.4	20.3	20.5	20.8	22.2	23.0	20.4	16.9	14.1
		MCWBR	14.1	15.2	11.2	9.7	8.8	9.3	10.7	11.4	10.9	10.4	10.2	12.0
Clear-Sky Solar Irradiance	taub	0.274	0.303	0.345	0.387	0.375	0.387	0.385	0.364	0.331	0.324	0.301	0.276	
	taud	2.159	2.112	2.107	2.138	2.249	2.296	2.322	2.351	2.398	2.323	2.213	2.141	
	Ebn at Noon	250	267	273	273	281	278	277	278	278	259	236	231	
	Edh at Noon	26	35	42	45	42	40	39	36	31	28	25	24	
All-Sky Solar Radiation	RadAvg	556	894	1348	1730	2080	2183	2107	1884	1487	902	516	416	
	RadStd	34	41	69	103	120	120	115	72	97	86	54	32	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	-1.01	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page